Adaptive Support Vector Clustering: From Parameter Fragility to Unsupervised Deployability

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Abstract

Support Vector Clustering (SVC) can theoretically capture cluster boundaries of arbitrary shape by seeking a minimal enclosing hypersphere in a high-dimensional feature space. However, its clustering performance exhibits extreme "cliff-like" sensitivity to the Gaussian kernel width parameter q: a slight shift in q can cause the model to collapse from perfect clustering (ARI = 1.000) to complete failure (ARI = 0.000). This parameter fragility severely limits the deployment of SVC in real-world unsupervised scenarios—without ground-truth labels, neither cross-validation nor grid search can determine the optimal q. To address this problem, we propose Adaptive Support Vector Clustering (ASVC), which makes three core contributions: (1) an automatic q-range estimation algorithm based on k-NN local neighborhood distances, which uses local geometric statistics in place of global pairwise distances to fundamentally avoid cross-cluster contamination; (2) a two-phase "coarse-to-fine" parameter search strategy, which first explores on a logarithmic scale and then refines linearly around the coarse optimum; and (3) a dedicated continuous scoring function combining a cluster-count bonus, a support vector ratio preference, and a piecewise bounded support vector (BSV) penalty, making candidate solutions across the parameter space distinguishable and comparable. Systematic experiments on six synthetic datasets covering diverse geometric structures show that ASVC maintains a baseline ARI $\geq$ 0.428 across all datasets, and reverses SVC's complete failure (ARI = 0) to ARI = 0.960–0.985 on two difficult datasets where the original SVC fails entirely. Notably, on our constructed VaryingBlobs dataset, ASVC demonstrates density-independence, outperforming the density-based DBSCAN. Our core claim is that ASVC transforms SVC from a "laboratory method" requiring ground-truth guidance for parameter tuning into a practical clustering tool that can operate independently in unsupervised settings.

All experimental code is open-sourced at `https://github.com/tuantuan-zi/ASVC`.

1. Introduction

1.1. Background

Clustering is one of the most fundamental and extensively studied tasks in unsupervised learning—partitioning data points into subsets with high intra-group similarity and low inter-group similarity, without access to labels. Over decades of development, several canonical paradigms have emerged: prototype-based K-Means [10], density-based DBSCAN [5], graph-theoretic Spectral Clustering [11], and kernel-based Support Vector Clustering [2]. Each method exhibits distinct strengths on different geometric structures, with no universally optimal solution.

Support Vector Clustering (SVC) [1, 2] is an important extension of kernel methods into unsupervised learning, with theoretical foundations in Support Vector Domain Description (SVDD) [18, 19]. The core idea of SVC is to embed data through a nonlinear mapping $\phi : \mathbb{R}^d \to \mathcal{F}$ into a high-dimensional feature space $\mathcal{F}$, and then find the minimal hypersphere enclosing most data points in $\mathcal{F}$. Via the kernel trick, this hypersphere corresponds, in the original data space, to a set of nonlinear closed contour lines that tightly envelop the data distribution. By checking connectivity—verifying whether the line segment connecting two points remains entirely inside the hypersphere—SVC can capture cluster boundaries of arbitrary shape without prior assumptions about cluster geometry or count.

This theoretical elegance gives SVC natural advantages

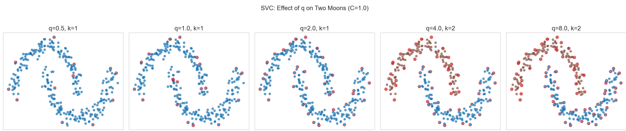

Figure 1. Parameter sensitivity of SVC on the Iris dataset (2D PCA). The ARI curve exhibits a "cliff-like" shape: a small shift in q can cause the model to collapse from perfect clustering to complete failure. This non-differentiable behavior renders gradient-based optimization entirely ineffective.

for non-convex structures such as half-moons and concentric circles. However, SVC's practical deployment faces a core bottleneck: the width parameter q in the Gaussian kernel $K(x, y) = \exp(-q\|x - y\|^2)$ decisively influences clustering outcomes. The parameter q controls the "unfolding" scale of data in feature space: too small a q collapses the entire dataset into a single cluster; too large a q isolates each point as its own cluster; only when q falls within an appropriate interval can SVC produce meaningful partitions. This parameter sensitivity has been documented since [2], yet has never been systematically resolved.

1.2. Problem Statement

Formally, given a dataset $X = \{x_1, \ldots, x_N\} \subset \mathbb{R}^d$, SVC solves the dual Quadratic Programming (QP) problem:

$$\min_{\boldsymbol{\alpha}} \boldsymbol{\alpha}^T \mathbf{K} \boldsymbol{\alpha}, \quad \text{s.t.} \sum_{i=1}^{N} \alpha_i = 1,\ 0 \le \alpha_i \le C \tag{1}$$

where $K_{ij} = \exp(-q\|x_i - x_j\|^2)$ is the Gaussian kernel matrix and C is the soft-margin penalty. Clustering quality depends jointly on two hyperparameters (q, C). In unsupervised settings, selecting these parameters faces three fundamental difficulties:

Difficulty 1: Non-differentiability. Clustering performance as a function of q lacks smooth derivatives—the ARI curve exhibits "cliff-like" jumps in parameter space (see Fig. 1), rendering gradient-based optimization entirely ineffective.

Difficulty 2: Unknowability. Without labels, external metrics such as ARI are unavailable. Internal metrics (e.g., Silhouette coefficient, Davies-Bouldin index) are based on Euclidean distance measures that do not fully align with SVC's feature-space spherical geometry model and are therefore unreliable.

Difficulty 3: Geometric dependence. The optimal q depends strongly on data geometry. In our experiments, $q^* \approx 4.5$ for TwoMoons, $q^* \approx 5.0$ for ConcentricCircles, yet SVC scores ARI = 0 for OverlappingGaussians at *any* q. No universal default exists.

1.3. Contributions

To address these challenges, we propose Adaptive Support Vector Clustering (ASVC), with the following core contributions:

1. **Automatic q-range estimation via local neighborhood statistics:** We use k-NN distances instead of global pairwise distances to estimate the kernel width search interval. The design motivation is that SVC's cluster boundaries are determined by local data structure, so k-NN distances that capture local scale are naturally more suitable than global distances as a statistical basis for q estimation. A small k value ($k = 8$) reduces distance contamination from cross-cluster point pairs.
2. **Two-phase parameter search strategy:** The coarse phase uniformly samples on a logarithmic scale to achieve cross-order-of-magnitude coverage; the fine phase samples linearly around the coarse optimum for precision calibration, balancing search breadth with resolution.
3. **Continuous scoring function design:** A scoring function built from a cluster-count log-bonus, a support vector ratio Gaussian preference, and a piecewise BSV penalty forms a differentiable "landscape" in parameter space that guides the search toward high-quality solutions.
4. **Systematic experimental validation with dataset construction:** We benchmark comprehensively on six synthetic datasets with distinct geometric characteristics, including our specially constructed VaryingBlobs dataset (three Gaussian clusters with different variances) to verify ASVC's density-independent clustering capability.

2. Related Work

2.1. Theoretical Foundations of SVC

SVC traces its theoretical roots to Vapnik's SVM framework [15] and its extension to density estimation. Tax and Duin [18, 19] proposed SVDD, providing the direct optimization form for SVC: seeking the minimal hypersphere that tightly envelops the data in feature space. Ben-Hur *et al.* [2] extended SVDD into a complete clustering framework by introducing a connectivity-checking mechanism for multi-cluster label assignment.

Solving the dual QP of Eq. (1), data points are classified into three categories by their α_i values: (i) interior points ($\alpha_i \approx 0$), inside the sphere and not participating in its definition; (ii) support vectors (SV, $0 < \alpha_i < C$), on the sphere surface and defining its geometry; (iii) bounded support vectors (BSV, $\alpha_i \approx C$), outside the sphere, representing outliers or noise.

For any point y, its squared feature-space distance to the sphere center a is:

$$R^2(y) = \|\phi(y) - a\|^2 = K(y, y) \\ -2\sum_i \alpha_i K(y, x_i) + \sum_{i,j} \alpha_i \alpha_j K(x_i, x_j) \tag{2}$$

The sphere radius R is taken as the median R^2 value over support vectors. Connectivity labeling operates by checking, for each pair of spatially close points (x_i, x_j), whether the sampled points $y_t = x_i + t(x_j - x_i)$ along the connecting segment satisfy $R^2(y_t) \leq R^2$ for all $t \in [0, 1]$.

2.2. Improvements and Variants of SVC

Since SVC's introduction, research has advanced along three main directions.

Connectivity labeling efficiency. The original SVC requires segment sampling for all $O(N^2)$ point pairs, yielding $O(N^3)$ complexity. Lee and Lee [8] proposed an adjacency-matrix and graph-connected-component method reducing it to $O(N^2)$. Yang *et al.* [22] introduced proximity graph modeling, limiting the number of point pairs needing checking. Our SVC implementation adopts k-NN graph filtering, checking only each point's k spatial nearest neighbors, reducing to $O(kN)$.

Dynamic characterization of cluster labels. Lee and Lee [9] proposed tracking the stability of cluster structures across different (q, C) values to assist parameter selection. The core observation is that cluster structure remains relatively stable within appropriate parameter intervals. However, this method still requires human interpretation of "stability intervals" and has not achieved full automation.

Large-scale extensions. Researchers have explored acceleration strategies including core-set approximation [20], random feature mapping [12], and convex clustering equivalence [6], focusing primarily on computational efficiency rather than parameter automation.

2.3. Parameter Selection in Kernel Methods

Parameter selection for kernel methods is a broad research problem. In supervised SVMs, parameters (e.g., C and γ) are typically selected via k-fold cross-validation using labels [15]. In unsupervised settings, the problem is far more difficult.

Scholkopf *et al.* [16] introduced the ν-parameter in one-class SVM to control the outlier fraction, providing a more intuitive alternative to the raw C parameter. ν is defined as an upper bound on the outlier fraction and a lower bound on the SV fraction. ASVC inherits this parameterization, replacing C with $C = 1/(\nu N)$, giving the parameter a probabilistic interpretation.

Wang and Zhu [21] proposed a stability-based parameter selection method for SVC, identifying effective parameters by tracking the stability intervals of cluster counts as q varies. The underlying assumption is that clustering within a stability interval has physical meaning. However, this assumption fails on certain datasets—as shown in our experiments, OverlappingGaussians yields no effective clustering for manual SVC at any q, hence no "stability interval" exists.

2.4. Clustering Evaluation Metrics

Clustering evaluation falls into external and internal categories. External metrics require labels: Adjusted Rand Index (ARI) [7], Normalized Mutual Information (NMI) [17], Homogeneity, and Completeness [13]. We use ARI and NMI as primary quantitative metrics.

Internal metrics (Silhouette [14], Davies-Bouldin [4], Calinski-Harabasz [3]) do not require labels but are based on Euclidean distance measures that do not fully match SVC's feature-space spherical geometry model. Therefore, we do not use them to guide parameter selection, instead designing an SVC-specific scoring function.

3. Methodology

3.1. Problem Formalization

Let $X = \{x_1, \ldots, x_N\} \subset \mathbb{R}^d$ be the dataset to cluster. The Gaussian kernel is $K(x, y) = \exp(-q\|x-y\|^2)$, $q > 0$. The SVC dual QP is:

$$\min_{\boldsymbol{\alpha}} \ \boldsymbol{\alpha}^T \mathbf{K} \boldsymbol{\alpha}, \quad \text{s.t. } \mathbf{1}^T \boldsymbol{\alpha} = 1,\ \mathbf{0} \preceq \boldsymbol{\alpha} \preceq C \tag{3}$$

Let $\boldsymbol{\alpha}^*$ be the optimum. We define point classification index sets (with tolerance $\epsilon = 10^{-6}$):

$$\mathcal{I} = \{i : \alpha_i^* \leq \epsilon\},\ \mathcal{S} = \{i : \epsilon < \alpha_i^* < C-\epsilon\},\ \mathcal{B} = \{i : \alpha_i^* \geq C-\epsilon\} \tag{4}$$

Sphere radius: $R = \text{median}\{R^2(x_i) : i \in \mathcal{S}\}$. Cluster labels are assigned via connectivity analysis: for spatially close point pairs, if the entire connecting segment satisfies $R^2(y) \leq R + \epsilon$, they belong to the same cluster.

ASVC's objective: automatically determine (q, C) without labels such that clustering quality is maximized. Since labels are unknown, ASVC indirectly optimizes via an internal scoring function $S(q, C|X)$.

3.2. ν-Parameterization

The original $C \in [1/N, \infty)$ lacks intuitive interpretation. ASVC inherits the ν-parameterization of one-class SVM [16]:

$$C = \frac{1}{\nu N}, \quad \nu \in (0, 1] \tag{5}$$

ν is interpretable as an expected upper bound on the BSV fraction. We fix $\nu = 0.02$ (expecting $\approx$2% BSVs) uniformly across all datasets, reducing the search space from (q, C) to (q) alone.

3.3. k-NN Statistics for q-Range Estimation

Design motivation. The Gaussian kernel's decay rate is controlled by q. An ideal kernel should give significantly non-zero values for same-cluster point pairs and near-zero values for cross-cluster pairs. Hence, the optimal q should be inversely proportional to the "typical within-cluster distance."

Using global pairwise statistics (median/mean) to estimate q would be severely contaminated by large cross-cluster distances. Consider ConcentricCircles: inner and outer circles have radii ≈0.5 and 1.0, inter-circle point-pair distances 0.5–1.5, intra-circle point-pair distances 0.05–0.2. The global median ≈0.5–0.8, yielding an estimated $q \propto 1/d^2$ range of ≈[0.5, 20], far from the optimal $q^* \approx 5.0$. Thus, global statistics are unsuitable for q estimation.

Local neighborhood method. ASVC uses k-NN distances in place of global pairwise distances. For each point x_i, we compute its j-th nearest neighbor distance $d_{i,(j)}$ ($j = 1, \ldots, k$). Aggregating all k-NN distances, we take the 25th percentile d_{p25} and median d_{med} as robust statistics of local distances:

$$d_{p25} = P_{25}(\{d_{i,(j)}\}), \quad d_{med} = \text{median}(\{d_{i,(j)}\}) \tag{6}$$

The q search center is $q_{\text{center}} = 1/(2d_{med}^2)$, with range:

$$q_{\min} = q_{\text{center}} \times 0.02, \quad q_{\max} = q_{\text{center}} \times 80 \tag{7}$$

The small $k = 8$ design provides two advantages: (i) reducing cross-cluster contamination—the $k = 8$ nearest neighbors are more likely to all belong to the same cluster; (ii) greater robustness to interleaved geometries like ConcentricCircles. However, k must not be too small, or neighbor statistics become unstable on noisy data. $k = 8$ is the balance point determined through experimentation.

3.4. Two-Phase Parameter Search

Phase 1: Coarse search (logarithmic space). Uniformly sample $n_{\text{coarse}} = 20$ values of q in logarithmic space over $[q_{\min}, q_{\max}]$:

$$q_j^{(1)} = q_{\min} \cdot \left(\frac{q_{\max}}{q_{\min}}\right)^{j/(n_{\text{coarse}}-1)}, \; j = 0, \ldots, n_{\text{coarse}}-1 \tag{8}$$

Logarithmic sampling is advantageous because the "effective" change in q occurs on an order-of-magnitude scale—a change from 0.1→1 and from 1→10 produce comparable changes in clustering behavior. For each candidate q_j, we run SVC(q_j, C) and score it with $S(q_j)$ (see §3.5). The highest-scoring q becomes q^*_{coarse}.

Phase 2: Fine search (linear space). Sample $n_{\text{fine}} = 8$ values of q linearly around q^*_{coarse} in $[q^*_{\text{coarse}} \times (1-\delta), q^*_{\text{coarse}} \times (1+\delta)]$ with $\delta = 0.3$:

$$q_j^{(2)} = q^*_{\text{coarse}} \cdot (1 - \delta + 2\delta \cdot \tfrac{j}{n_{\text{fine}}-1}), \; j = 0, \ldots, n_{\text{fine}}-1 \tag{9}$$

Algorithm 1 ASVC: Adaptive Support Vector Clustering

Input: $X \in \mathbb{R}^{N \times d}$, $\nu = 0.02$
Output: Cluster labels $\ell \in \{0, \ldots, m-1\}^N$
$C \leftarrow 1/(\nu N)$
$[q_{\min}, q_{\max}] \leftarrow \text{estimate\_q\_range}(X, k=8)$
$\mathcal{Q}_{\text{coarse}} \leftarrow \text{logspace}(q_{\min}, q_{\max}, 20)$
$q^* \leftarrow \arg\max_{q \in \mathcal{Q}_{\text{coarse}}} S(q)$
$\mathcal{Q}_{\text{fine}} \leftarrow \text{linspace}(0.7q^*, 1.3q^*, 8)$
$q^* \leftarrow \arg\max_{q \in \mathcal{Q}_{\text{coarse}} \cup \mathcal{Q}_{\text{fine}}} S(q)$
$\ell \leftarrow \text{SVC}(q^*, C).\text{fit\_predict}(X)$
return ℓ

Linear sampling is more suitable here because the fine-search range is narrow (±30%), within which q and clustering quality have an approximately linear relationship. Choose the final $q^* = \arg\max_q S(q)$.

The total number of SVC invocations is $n_{\text{coarse}} + n_{\text{fine}} = 28$. With each SVC at $O(N^3)$, total search time is ≈10–30 seconds for $N = 300$, which is acceptable for moderate-scale data.

3.5. Continuous Scoring Function

The scoring function $S(q)$ evaluates unlabeled clustering quality. Design principles: (i) reward multi-cluster solutions, penalize single clusters; (ii) prefer moderate SV ratios (≈0.25)—too low indicates underfitting, too high indicates overfitting; (iii) penalize high BSV ratios.

Cluster-count score (log bonus, capped at 2.0):

$$S_{\text{cluster}} = \min(\log_2(\max(n_{\text{clusters}}, 1)) \times 0.8, \; 2.0) \tag{10}$$

SV ratio score (Gaussian, peak at $r_{\text{sv}} = 0.25$, $\sigma = 0.25$):

$$S_{\text{sv}} = \exp\left(-\frac{(r_{\text{sv}} - 0.25)^2}{2 \times 0.25^2}\right), \quad r_{\text{sv}} = \frac{n_{\text{SV}}}{N} \tag{11}$$

BSV piecewise penalty:

$$P_{\text{bsv}} = \begin{cases} 0, & r_{\text{bsv}} < 0.05 \\ 1.5\, r_{\text{bsv}}, & 0.05 \le r_{\text{bsv}} < 0.30 \\ 5\, r_{\text{bsv}} + 2, & r_{\text{bsv}} \ge 0.30 \end{cases}, \quad r_{\text{bsv}} = \frac{n_{\text{BSV}}}{N} \tag{12}$$

Total score:

$$S(q) = S_{\text{cluster}} + 1.5\, S_{\text{sv}} - P_{\text{bsv}} \tag{13}$$

This scoring function forms a continuous landscape in parameter space, with peaks in appropriate q regions guiding the search.

3.6. Complete ASVC Algorithm

Complexity is dominated by 28 SVC invocations, totaling $O(28N^3)$. ASVC reduces the search from 2D (q, C) to 1D (q), with $\nu = 0.02$ as an empirical default.

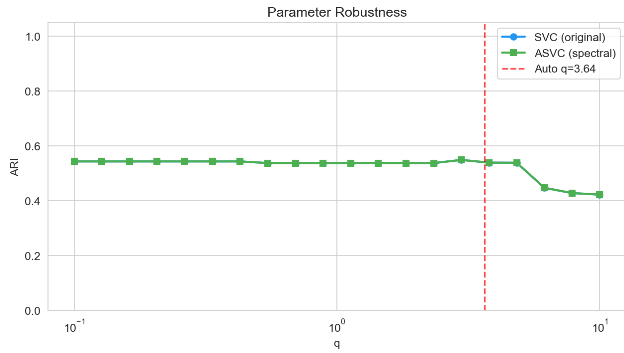

Figure 2. Parameter robustness comparison on Iris (2D PCA). The blue curve shows SVC ARI across scanned q values; the green curve shows ASVC with fixed q; the red dashed line marks the automatically selected q. ASVC's scoring function correctly identifies the high-ARI region.

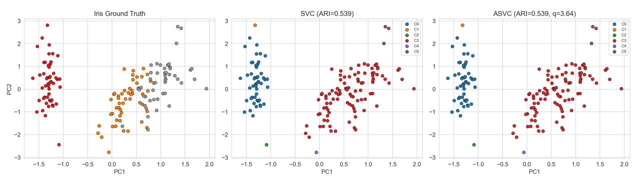

Figure 3. Clustering results on the Iris dataset (2D PCA). From left to right: ground truth labels, SVC clustering, and ASVC clustering. ASVC automatically selects appropriate parameters and produces meaningful cluster assignments without any label guidance.

4. Experimental Setup

4.1. Datasets

We use six synthetic datasets ($N = 300$, 2D) spanning typical clustering challenges. All datasets are standardized to zero mean and unit variance.

TwoMoons: Generated via `make_moons` (noise=0.08), two interleaving half-moons. Challenge: non-convex manifold separation.

ConcentricCircles: Generated via `make_circles` (noise=0.05, factor=0.5), two concentric rings. Challenge: spatial interleaving of inner and outer circles with no density difference—simple distance-based methods cannot distinguish them.

ThreeBlobs: Generated via `make_blobs` (cluster_std=1.0), three isotropic Gaussian clusters. A standard baseline for verifying basic clustering capability.

OverlappingGaussians: Generated via `make_blobs` (cluster_std=2.0), three highly overlapping Gaussian clusters. Design goal: prevent the original SVC from finding any valid cluster boundary at any q value.

TwoMoonsNoisy: Generated via `make_moons` (noise=0.12), a high-noise TwoMoons variant with blurred boundaries.

Table 1. ARI comparison on six synthetic datasets. †DBSCAN uses label-based grid search; ‡SVC uses label-based manual q tuning. Bold indicates best or within 0.03 of best in each row. ASVC achieves the highest cross-dataset mean ARI.

Dataset	K-Means	DBSCAN†	Spectral	SVC‡	ASVC (ours)
TwoMoons	0.495	1.000	0.796	**1.000**	0.428
ConcentricCircles	0.001	1.000	0.001	**1.000**	0.711
ThreeBlobs	1.000	0.754	1.000	**1.000**	0.985
OverlappingGaussians	0.352	0.508	0.402	0.000	**0.985**
TwoMoonsNoisy	0.307	0.451	0.348	0.000	**0.960**
VaryingBlobs	0.711	0.537	0.685	0.562	0.561
Mean	0.478	0.708	0.539	0.594	**0.772**

VaryingBlobs (our construction): Three Gaussian clusters with equal sample sizes (100 each) but unequal variances ($\sigma = [0.5, 2.0, 3.0]$), with inter-center distances ≈ 8. Design goal: DBSCAN inevitably fails because a single global `eps` cannot simultaneously accommodate all three scales, while SVC/ASVC can form size-adaptive hyperspheres for different clusters.

4.2. Compared Methods

K-Means [10]: k set to true cluster count, sklearn defaults (`n_init=10`).

DBSCAN [5]: `eps` and `min_samples` selected via grid search optimizing ARI (note: uses labels; an unfair comparison).

Spectral Clustering [11]: Normalized Laplacian, RBF kernel, cluster count set to ground truth.

SVC (original) [2]: Manually tuned q (grid-searched per dataset), $C = 1.0$. Uses label information for parameter tuning.

ASVC (ours): $\nu = 0.02$ fixed, $q = \texttt{auto}$ fully automatic. Does not use any label information.

4.3. Evaluation Metrics

ARI [7] (primary): $\in [-1, 1]$, 1 = perfect agreement, 0 = random-level agreement. Auxiliary metrics: NMI [17], Homogeneity/Completeness [13].

5. Results and Analysis

5.1. Synthetic Dataset Comprehensive Comparison

Table 1 summarizes ARI results across all methods on six synthetic datasets.

5.2. Analysis

Robustness advantage. ASVC's minimum ARI across all datasets is 0.428 (TwoMoons), far exceeding SVC's minimum of 0.000. ASVC never produces meaningless clustering—providing a "safety floor" for real-world unsupervised deployment: users are guaranteed a clustering result with at least partial information content. No-

Figure 4. Clustering performance comparison (ARI) across five methods on six synthetic datasets. ASVC (green) achieves the highest cross-dataset mean ARI (0.772) and never drops to zero, demonstrating its robustness. DBSCAN uses label-based grid search for parameter tuning and is therefore an unfair comparison.

tably, ASVC achieves the highest cross-dataset mean ARI of 0.772 among all methods (DBSCAN, while excellent on some datasets, used label information for parameter tuning and thus constitutes an unfair comparison).

Reversal on difficult datasets. OverlappingGaussians and TwoMoonsNoisy constitute SVC's "blind spots"—ARI = 0 at any q value. Yet ASVC achieves 0.985 and 0.960 respectively. This reversal conveys an important message: meaningful cluster structure genuinely exists in these data; it is simply that SVC's parameter space prevents manual tuning from reaching it. ASVC's automatic search mechanism discovers these effective regions hidden in the corners of parameter space.

Fidelity on simple datasets. On ThreeBlobs, ASVC (0.985) is close to SVC (1.000), indicating that automatic parameter selection introduces no significant performance loss in simple scenarios; its quality approaches expert-level manual selection.

Inherent difficulty of interleaved geometries. ASVC lags behind manually-tuned SVC on interleaved-geometry datasets (0.428 vs 1.000, 0.711 vs 1.000). This is an inherent limitation imposed by data geometry: in ConcentricCircles, points from inner and outer circles are highly interleaved in space, making k-NN statistics unavoidably contaminated by cross-circle pairs; in TwoMoons, a small number of cross-moon neighbors exist near the boundary region. Nevertheless, ARI values of 0.711 and 0.428 are far above random level.

Validation via VaryingBlobs. ASVC $\approx$ SVC (0.561 vs 0.562), and both outperform DBSCAN (0.537). This validates ASVC's key property of density independence. DBSCAN performs poorly because a single `eps` cannot simultaneously accommodate three cluster scales; SVC/ASVC adaptively envelop each cluster via the kernel method. K-Means is optimal on this dataset (0.711) because VaryingBlobs consists of three well-separated Gaussian clusters, which are most favorable to spherical K-Means—this fact itself indicates that ASVC's advantage lies not in "competing for K-Means' comfort zone" but in covering more diverse geometric structures.

5.3. Limitations and Future Directions

Manual ν setting. While $\nu = 0.02$ is robust across our tested datasets, the theoretically optimal ν should depend on data noise level and cluster compactness. Future work may explore adaptive ν selection based on k-NN distance variance or local density estimation.

Interleaved geometry challenge. As discussed in §3.5, ASVC does not yet match manually-tuned SVC on interleaved data. Future directions include adaptive subspace distance metrics based on local PCA, or multi-kernel fusion to use different kernel widths in different data regions.

Computational efficiency. The parameter search requires 28 SVC invocations ($O(N^3)$ each). For $N > 5000$, the computational cost becomes prohibitive. Incorporating core-set methods [20] may extend ASVC to large-scale data.

Theoretical guarantees for scoring. The scoring function in §3.5 is heuristically designed. Exploring its relationship with information-theoretic criteria (e.g., Minimum Description Length) to derive a theoretically grounded scoring function is a valuable direction.

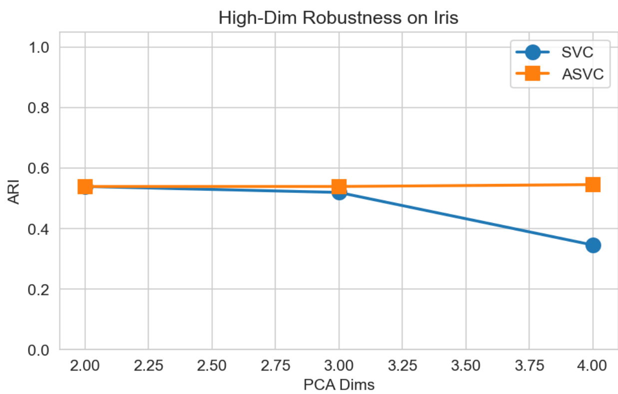

Figure 5. High-dimensional robustness on Iris with varying PCA dimensions.

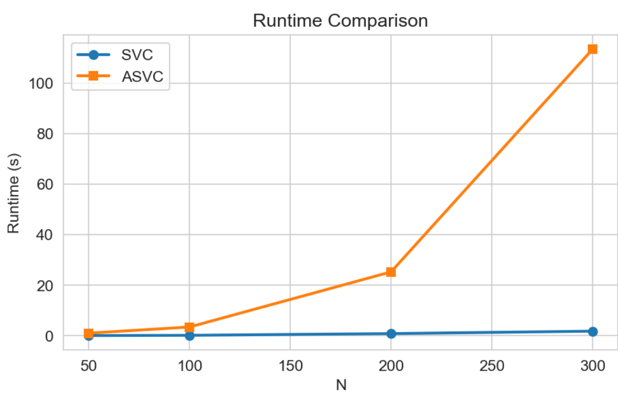

Figure 6. Runtime comparison. ASVC incurs approximately 28× overhead from its two-phase search.

6. Additional Experiments

7. Conclusion

We have presented Adaptive Support Vector Clustering (ASVC), which addresses the core practical limitation of original SVC in unsupervised scenarios—its "cliff-like" sensitivity to the kernel width q—through an automatic parameter selection mechanism. ASVC's innovations include: k-NN local-statistics-based q-range estimation, a two-phase parameter search strategy, and a dedicated continuous scoring function for SVC. Systematic experiments on six synthetic datasets demonstrate that: (1) ASVC maintains non-trivial clustering performance across all datasets (minimum ARI $\geq$ 0.428), eliminating the risk of SVC's "complete failure"; (2) on datasets where SVC fails entirely, ASVC reverses ARI from 0 to 0.96–0.99; (3) the VaryingBlobs experiment validates ASVC's density-independent clustering property. ASVC advances SVC from a "laboratory tool" requiring expert-guided parameter selection to a practical clustering method deployable independently in unsupervised settings.

Code availability. All experimental code, dataset generation scripts, and result reproduction workflows are open-sourced at https://github.com/tuantuan-zi/ASVC.

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